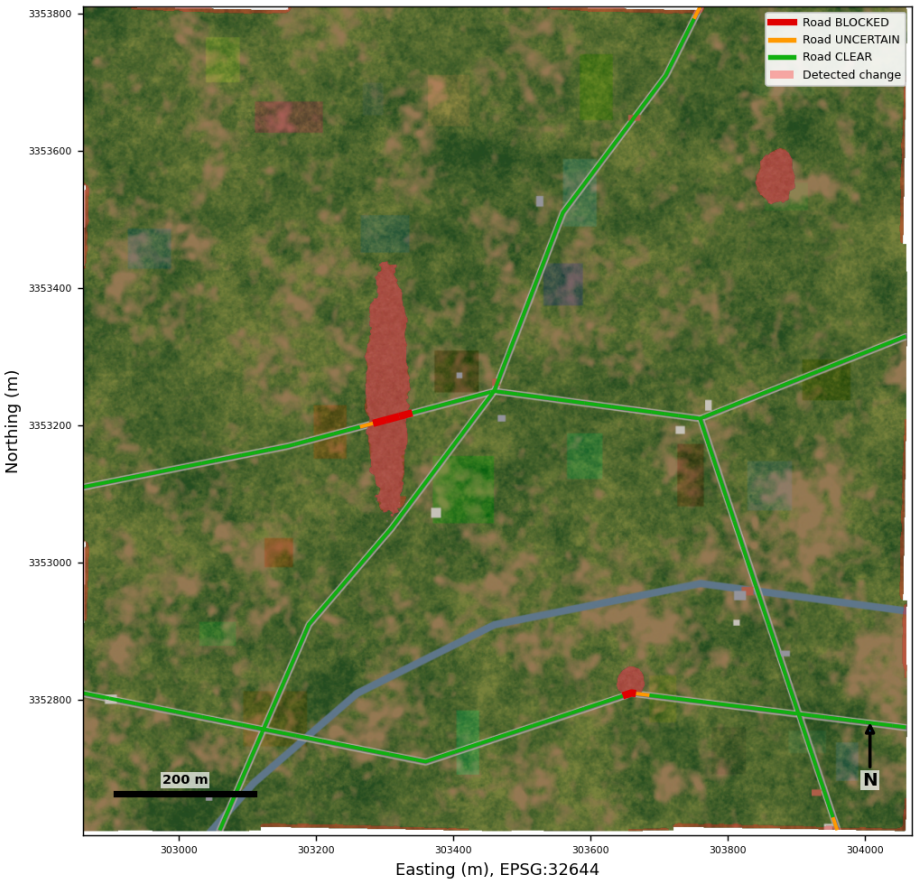
MAPPA: rapid post-landslide map brief

SIMULATED DEMONSTRATION DATA. Imagery, roads and the landslide are synthetic. Not for operational use.



Area mapped	1.434 km2 at 1 m/px
Changed area	33,651 m2 in 13 patches
Roads blocked	79 m
Roads uncertain	79 m
Roads clear	4345 m

Blocked stretch (road)	Length m	Changed
NH-7 Main Road	59	88%
Hill Track	20	89%

Accuracy statement. Positional accuracy +/-10 m (95%), unverified against ground truth. Relative accuracy inside the map is much better (centimetre-to-decimetre class) but was not measured here. Method: SIFT+RANSAC similarity (409 inliers, registration RMS 1.44 m). GPS-only georeference was off by about 5.88 m before correction. Absolute accuracy is limited by the base layer (assumed +/-10 m, 95%).

Synthetic validation (known truth). Injected GPS-only error averaged 5.69 m (max 9.64 m). After registration the mean error is 0.14 m (max 0.26 m) over 25 check points. Road check vs. true slide footprint: precision 1.0, recall 1.0. This is a synthetic test, not a field accuracy claim.

Change detection limits. Detection is coarse: limited by the 9 m base imagery; small debris is not resolved. Height check used (DSM minus DEM, 12.42 m vertical bias removed).

Road rule. 20 m segments, 5 m buffer. BLOCKED: changed share >= 40% and height anomaly >= 0.5 m (without height data: >= 60%). UNCERTAIN: changed share >= 15%. Thresholds are tunable and not field-calibrated.

Processing. 1.43 km2 analysed at 1 m/px, 1 CPU core(s), 3.9 GB RAM, no GPU: 5 s (this pipeline only; photogrammetry stitching time is measured separately).

Provenance. Synthetic scene generated by mappa.demo_data with a known injected GPS error. Built entirely with open-source tools (OpenDroneMap, OpenCV, rasterio, GeoPandas, Shapely, Matplotlib, ReportLab). Fully offline. Open GeoPackage / GeoJSON / KML / GeoTIFF in QGIS.